The Strong Force as a Repulsive Layer of Proximity Gravity: Proton Chaos, Magic Numbers, Slow Neutrons, and Cold Fusion

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Abstract

Proximity gravity is the gravitational channel of double spacetime theory at a separation comparable to its characteristic length ℓ. On the equal-scale locus the force is the Newtonian force divided by the interference factor γ_s. The sign of γ_s is the repulsion. Masses attract where $\gamma_s > 0$ and repel where $\gamma_s < 0$. The first negative interval is the repulsive inversion layer. Its outer edge is the first node x_1, the unique root of $\tan x = \coth x$ in $(\pi/4, \pi/2)$, and the numerical values are $x_1 = 0.93755$, $r_1/\ell = 0.53330$. The inner edge is the second node, at $r_2/\ell = 0.12731$. Both sides of the first node drive the separation toward the node, and on the equal-scale locus that drive does not carry the separation through the layer into the deeper attractive chamber. The length is the charged-pion Compton wavelength $\lambda_\pi \approx 1.41\,\mathrm{fm}$. The laboratory signs place it strictly between $56/75\,\mathrm{fm}$ and $19/10\,\mathrm{fm}$: repulsion at $\frac{2}{5}\,\mathrm{fm}$ forces $\ell > 56/75\,\mathrm{fm}$, and attraction at $1\,\mathrm{fm}$ forces $\ell < 19/10\,\mathrm{fm}$. At that length the first node lies strictly between $14/19$ and $213/280\,\mathrm{fm}$, and the rapidity wall lies strictly between $8/9$ and $142/157\,\mathrm{fm}$.

A relative rapidity φ, with $\tanh\varphi = |v|/c$, is added to the usual proximity rapidity while the dual angle is held at ℓ/r. The interference factor becomes $\cosh(\varphi/2)\,\gamma_s - \sinh(\varphi/2)\,\sigma$. Inside the window $r_1 < r < 2\ell/\pi$ the factor changes sign at the closed threshold

$$\varphi_*(x) = \ln\frac{\sin x + \cos x}{\sin x - \cos x} - 2x, \qquad x = \ell/(2r).$$

Below the threshold the channel attracts, and the attraction strengthens as the rapidity rises. Above it the channel repels, and the outward force is arbitrarily large immediately above the threshold. Outside the radius $2\ell/\pi$ no finite rapidity produces repulsion. The outer radius of the inversion therefore increases with φ and remains strictly below $2\ell/\pi$, which is 1.1937 times the static node. At small rapidity the node moves by $\varphi/(4\cosh^2 x_1)$ in the phase, and the coefficient is 0.11528. A slow neutron remains attractive down to a skin of that thickness on the static node. A nonnegative rapidity that increases while it stays below the threshold meets the threshold at one phase strictly below x_1. The free infall is of that kind for as long as the channel attracts, so it meets the threshold before the static node. Reaching the node while the channel is still attractive requires the rapidity to fall. A neutron of kinetic energy $1\,\mathrm{MeV}$, $10\,\mathrm{MeV}$, and $100\,\mathrm{MeV}$ meets the threshold at $1.0056\,r_1$, $1.0175\,r_1$, and $1.0517\,r_1$. That threshold is an entrance condition, common to capture, to neutron-induced fission, and to fusion: above it the nuclear channel already repels, and none of the three can begin. It is not the split. The disposal of rapidity inside a compound nucleus is not derived, and neither is the laboratory fall of a cross section as $1/v$.

Two protons carry the electromagnetic channel as well. With the nuclear prefactor written κ, the separation force is $-\kappa/(\gamma\, r^2) + ke^2/(\gamma_E r^2)$. Gravitational nodes at which γ_s falls through zero attract the separation; electromagnetic nodes of the same kind repel it. Consecutive nodes of one channel have opposite stability. When the two lengths differ, the nodes interleave, so attraction and repulsion occupy neighbouring chambers. In a nucleus different pairs occupy different chambers at one instant. The force on a proton is unbounded on the wall of a chamber, and an arbitrarily small displacement that puts one pair on a node changes that force by an arbitrarily large amount. That sensitivity is the chaotic proton force. The identification $\kappa = Gm_p^2$ loses to the Coulomb term unless $|\gamma|$ compensates a factor 1.236×10^{36}. The MeV depth is an input.

For a charged pair the same threshold decides the contest with the Coulomb term, wherever the electromagnetic factor stays positive. Above the threshold the nuclear term is outward, and the net repulsion exceeds the Coulomb repulsion. Below it the nuclear term is inward, and on an interval of rapidity that ends at the threshold it exceeds every finite ratio of the Coulomb prefactor to κ. A lower rapidity moves that onset closer to the static node. When the kinetic energy is at most 10^{-10} of the rest energy, the fractional widening of the radius is less than 5×10^{-6}, and a nucleon or a deuteron at room temperature is of this kind. Outside $2\ell/\pi$ a large rapidity drives the nuclear attraction to

zero and leaves the Coulomb repulsion. The free infall still meets the threshold before the node, so the slow charged entrance is a path on which the rapidity is held down. The host of that holding, along one axis, is the restoring seat outside the first charge node in a regular octahedron of positive ions [3]. The rate is not derived.

The integers 2, 8, 20, 28, 50, 82, and 126 are seat counts. The spin-weighted oscillator closure through shell N is $C(N) = (N+1)(N+2)(N+3)/3$. For $N \geq 2$ the further closure $C(N)+2(N+2)$ is that count plus the degeneracy of the highest-j orbital of the next shell. These integers are not indices of the nodes. A doubly magic nuclide closes one such count for each rotor class. The layer geometry does not select the integers, and it does not fix the valley of stability.

A dual excitation is a different coordinate from the relative rapidity. At fixed excitation $\psi \in (0, \pi)$ the outer well has a single attractor, at the phase where the critical angle $2\arctan(\gamma_s/\tau)$ equals ψ. That critical angle falls strictly from π to 0 between infinity and the first node, and its deficit from π begins as $2\ell/r$. Relative rapidity cannot invert the force outside $2\ell/\pi$. Dual excitation can, at every finite radius. Fission, once a compound nucleus exists, is that inversion. Fusion is the entrance that stays inward. A slow neutron can open fission only by passing the rapidity threshold first.

1 Introduction

The nuclear section of double spacetime theory reads the strong force as proximity gravity: the gravitational channel on the equal-scale locus, where the interference factor γ_s changes sign on a discrete set of radii [1]. No second coupling is introduced. The same section is explicit about what that reading does not do. The Newtonian combination Gm_p^2 is not an $O(1)$ nuclear strength. Near a node, $1/\gamma_s$ diverges, and that divergence is a kinematic amplification of a channel already present. The femtometer and MeV scales enter through a length ℓ and through the masses, and ℓ is not derived. Magic numbers are named as resonance conditions and are not extracted from the nodes. Fission and fusion are named as layer transitions.

This note computes the layer that those statements point to. Section 2 fixes the signed force and the two auxiliary functions that the later thresholds are built from. Section 3 is the repulsive inversion itself: the first interval on which γ_s is negative, the direction of the force on each side of its walls, and the sense in which the equal-scale force drives a separation toward the outer wall without carrying it through the layer. Section 4 puts a relative rapidity into the usual sector only. The threshold at which the force becomes repulsive is a closed function of the phase, and it exists only between the static node and the radius $2\ell/\pi$. A free inward rapidity meets that threshold before the static node. Capture, neutron-induced fission, and fusion share that entrance. A slow neutron is a neutron that has not crossed it. Section 5 adds the electromagnetic channel for a proton pair. The two channels do not share their walls unless their lengths are equal, and a nucleus of several protons therefore holds attracting pairs and repelling pairs at once. Section 6 is the slow charged entrance. Below the rapidity threshold the nuclear term opposes the Coulomb term and, next to the threshold, exceeds it; above the threshold the nuclear term adds to the Coulomb repulsion. A thermal rapidity leaves that onset against the static node. Section 7 separates the seat counts 2, 8, 20, 28, 50, 82, and 126 from the node sequence, and lists the doubly closed nuclides. Section 8 returns to a dual excitation at fixed separation. That excitation, unlike the rapidity, can invert the outer well at every radius. The inversion is the split of a nucleus already formed. It is not a second way in.

The electromagnetic companion has already used the same γ_s for charges [2]. The signs below are the gravitational specialisation of that force, together with the electromagnetic term where the particles are protons. The discrete companion represents a nucleus as an ordered product of proton and neutron rotor classes and proves that each fixed compactification admits only finitely many algebraically stable mass numbers [4]. The numerical ceiling near $A \sim 300$ remains the heuristic already stated there. Nothing in the layer geometry of this note replaces it.

2 The signed proximity force

On one axis the interference factor of the dual rotor is

$$\gamma_s(\alpha, \beta) = \cosh\frac{\alpha}{2}\cos\frac{\beta}{2} - \sinh\frac{\alpha}{2}\sin\frac{\beta}{2}. \tag{1}$$

The equal-scale locus sets $\alpha = \beta = \ell/r$. In the dimensionless phase $x = \ell/(2r)$,

$$\gamma_s(x) = \cosh x \cos x - \sinh x \sin x, \qquad \gamma_s'(x) = -2\cosh x \sin x. \tag{2}$$

Thus $\gamma_s(0) = 1$. The Taylor expansion used below is

$$\gamma_s(x) = 1 - x^2 - \frac{1}{6}x^4 + O(x^6). \tag{3}$$

Two further combinations enter the thresholds. They are

$$\sigma(x) = \cosh x \sin x - \sinh x \cos x, \tag{4}$$
$$\tau(x) = \cosh x \sin x + \sinh x \cos x. \tag{5}$$

Both vanish at the origin. Their derivatives are $\sigma'(x) = 2 \sinh x \sin x$ and $\tau'(x) = 2 \cosh x \cos x$. The sum and difference collapse to

$$\sigma + \gamma_s = e^{-x}(\sin x + \cos x), \qquad \sigma - \gamma_s = e^{x}(\sin x - \cos x). \tag{6}$$

On $(0, \pi)$ one has $\sigma(x) > 0$, because $\sigma(0) = 0$ and σ' is nonnegative on that interval and not identically zero on any subinterval. On $(0, \pi/2)$ one has $\tau(x) > 0$, because $\tau(0) = 0$ and $\tau' > 0$ there.

The force that increases the separation of two nucleon masses is

$$F_r = -\frac{\kappa}{\gamma\, r^2}. \tag{7}$$

Here γ is (1) evaluated on the rapidities assigned to the pair, and $\kappa > 0$ is the prefactor of the nuclear channel. Where $\gamma > 0$ the force is inward. Where $\gamma < 0$ it is outward. On the equal-scale locus, $\gamma = \gamma_s(x)$. The main text writes the same quotient with the Newtonian prefactor Gm_1m_2 [1]. That identification is not used as the nuclear well. Section 5 records the size of the discrepancy with the Coulomb prefactor. The radii, the sign, and both thresholds are independent of κ.

At a simple zero of γ the force has a simple pole. The proper-time factor of the parent theory remains positive at that zero: a node is a divergence of the force, not a reversal of the clock [1].

3 The repulsive inversion layer

On each interval $[n\pi, (n+1)\pi]$ the derivative (2) has the sign of $-\sin x$, so γ_s is strictly monotone there and has exactly one zero. The zero lies where $\tan x = \coth x$. Since $\coth x > 1$ for finite positive x, the n-th zero sits in

$$n\pi + \frac{\pi}{4} < x_n < n\pi + \frac{\pi}{2}.$$

The first zero x_1 is the unique root in $(\pi/4, \pi/2)$. Numerically $x_1 = 0.93755$, so

$$\frac{r_1}{\ell} = \frac{1}{2x_1} = 0.53330.$$

The second zero is $x_2 = 3.9274$, and $r_2/\ell = 0.12731$. The ratio of those radii is $r_2/r_1 = x_1/x_2 = 0.23872$. The same bounds as in the main text,

$$\frac{1}{6} < \frac{r_2}{r_1} < \frac{4}{5\pi},$$

follow from the bracket $\pi/4 < x_1 < 1$, together with $x_2 \in (5\pi/4, 3\pi/2)$. The upper estimate is $x_1/x_2 < 1/(5\pi/4) = 4/(5\pi)$. The lower estimate is $x_1/x_2 > (\pi/4)/(3\pi/2) = 1/6$. Ordered by decreasing radius, $\gamma_s > 0$ for $r > r_1$, $\gamma_s < 0$ on $r_2 < r < r_1$, and the sign continues to alternate. The interval $r_2 < r < r_1$ is the repulsive inversion layer.

The extremum of γ_s inside that layer is $\gamma_s(\pi) = -\cosh \pi$, at radius $\ell/(2\pi)$. The outward force there is

$$|F_r| = \frac{4\pi^2}{\cosh \pi}\, \frac{\kappa}{\ell^2} = 3.4057\, \frac{\kappa}{\ell^2}. \tag{8}$$

Plateaux of higher index are quieter. At $x = n\pi$ the same reduction gives

$$|F_r| = \frac{4\pi^2 n^2}{\cosh(n\pi)}\, \frac{\kappa}{\ell^2},$$

which falls geometrically because $\cosh(n\pi)$ grows as $e^{n\pi}/2$. What grows without bound toward the centre is the set of poles, not the force between them.

The first node is an attractor of the separation. As x approaches x_1 from below, $\gamma_s \downarrow 0$ and $F_r \to -\infty$, so the separation decreases and the phase increases toward the node. As x approaches x_1 from above, $\gamma_s \uparrow 0$ through negative values and $F_r \to +\infty$, so the separation increases and the phase falls back toward the node. The same pairing of signs holds at every zero where γ_s' is negative, that is, on every interval $(2m\pi, (2m+1)\pi)$. On the complementary intervals γ_s' is positive, the crossing runs from negative values to positive values, and both sides of the zero drive the separation away from it. The second node is of that second kind. Inside the inversion layer the outward force therefore points toward r_1, and just inside r_2 the inward force points away from r_2, deeper into the next attractive chamber. On the equal-scale locus a separation that starts large is driven toward the first node, and not through the layer into the chamber beyond r_2. Once the rapidity of the infall is allowed to grow, the path leaves that locus. It meets the threshold of Section 4 at a phase strictly below x_1, and the deeper chamber remains unreached.

The radial integral of (7) on the outer side of the node falls without a lower bound, by the same substitution that the electromagnetic companion uses for the Coulomb well [2]. The path from infinity does not cross a pole, and the integral diverges as the upper phase approaches x_1. The inversion layer is on the other side of that pole. Its local force is defined wherever $\gamma_s \neq 0$, and the outer-well integral does not continue across the node.

The length is the charged-pion Compton wavelength $\ell = \lambda_\pi \approx 1.41\,\mathrm{fm}$. The main text records that every equal-scale node then lies below 1 fm, with the outermost node confined to the interval 0.44–0.91 fm [1]. The central value is $r_1 = 0.752\,\mathrm{fm}$. The identification is not a derivation of ℓ. Placing the inversion layer at 1–2 fm requires a longer length. The ratios of the node radii are independent of the length.

The signs of the nucleon–nucleon interaction place this length in a window. Realistic potentials fitted to the scattering data are repulsive for separations up to $\frac{1}{2}$ fm and attractive from 1 fm to 2 fm [6]. A separation of $\frac{2}{5}$ fm lies inside that core. For a slow pair, repulsion there places the separation inside the first node. The node lies strictly above 14/15, so

$$\ell > \frac{28}{15} \cdot \frac{2}{5} = \frac{56}{75}\,\mathrm{fm}.$$

Attraction at 1 fm places that separation in the outer well. The node lies strictly below 19/20, so $\ell < (19/10)\,\mathrm{fm}$. The two radii differ by a factor of two. A widening of the node by one fifth, larger than the five percent already found at 100 MeV, still leaves both $\frac{2}{5}$ fm and $\frac{1}{2}$ fm inside 1 fm. The charged-pion wavelength lies strictly between 56/75 fm and 19/10 fm.

At this length the first node lies strictly between 14/19 and 213/280 fm, hence strictly above $\frac{1}{2}$ fm and strictly below 1 fm. The second node lies strictly below $\frac{1}{2}$ fm, so the core scale lies in the inversion layer. The rapidity wall $2\ell/\pi$ lies strictly between 8/9 and 142/157 fm, hence strictly below 1 fm. No finite rapidity inverts the force at the inner edge of the attractive interval.

The measured charge radii are sizes of charge distributions. They are not pair separations. Table 1 records the rms values of Angeli and Marinova, together with the quoted uncertainties [7]. Each lower edge, the central value minus that uncertainty, exceeds 1 fm, so at this length each lies outside the rapidity wall.

Table 1: Root-mean-square charge radii. A charge radius is not a pair separation. Each lower edge lies outside the rapidity wall under $\ell = \lambda_\pi$.

Nuclide	R (fm)	uncertainty (fm)
^{2}H	2.1421	0.0088
^{4}He	1.6755	0.0028
^{16}O	2.6991	0.0052
^{40}Ca	3.4776	0.0019
^{48}Ca	3.4771	0.0020
^{208}Pb	5.5012	0.0013

The mean volume per nucleon at the saturation density $n_0 = 0.16\,\mathrm{fm}^{-3}$ is $25/4\,\mathrm{fm}^3$. Its cube root lies strictly between 46/25 and 37/20 fm. That spacing lies outside the rapidity wall. The cap it gives is $\ell < (19/10) \cdot (37/20)\,\mathrm{fm}$, which is weaker than the cap from attraction at 1 fm.

The parent theory reads the exclusion of a shared seat as this layer, rather than as a separate statistical postulate: identical fermions that overlap meet the outward side of the first node [1]. That reading is adopted here for nucleons. It is not a derivation of the spin-statistics theorem. A proton and a neutron are different rotor classes [4], so the same sentence does not by itself forbid a proton–neutron pair at one separation.

The binding scale is not in the geometry. The main text quotes $\mathrm{BE}/A \sim 8\,\mathrm{MeV}$ near iron, and an optical-model depth of order 40–50 MeV, as distinct inputs. Both remain inputs. Saturation of the density, in the form $R = r_0 A^{1/3}$, is the regularity the layer picture claims, and the main text already records that the two common numerical conventions for r_0 disagree at the level of ten percent. This note does not choose between them.

4 Relative rapidity

Relative motion is a usual-sector boost. In the parent theory the rapidity of that boost is the positive parameter φ with $\tanh\varphi = |v|/c$, the square root of the sum of squares of the rapidity components [1]. The reading adopted here adds that parameter to the proximity value and leaves the dual angle at the proximity value,

$$\alpha = \frac{\ell}{r} + \varphi, \qquad \beta = \frac{\ell}{r}, \qquad \varphi \geq 0. \tag{9}$$

Equal scale is broken as soon as $\varphi \neq 0$. The pair leaves the locus on which the mismatch scalar J vanishes. The sign of the force is still the sign of (1). The inverse boost, $\alpha \mapsto -\alpha$, is a different element: it changes the sign in front of $\sinh(\alpha/2)$ and raises the interference factor. Capture is read on the positive rapidity, which is the parameter the rotor actually carries.

Write $\delta = \varphi/2$. Substitution into (1) gives

$$\gamma(x,\delta) = \cosh\delta\,\gamma_s(x) - \sinh\delta\,\sigma(x). \tag{10}$$

The closed form

$$\gamma(x,\delta) = \frac{1}{2}e^{\delta+x}(\cos x - \sin x) + \frac{1}{2}e^{-(\delta+x)}(\cos x + \sin x) \tag{11}$$

follows from (6), and it makes the sign plain. For $0 \leq x \leq \pi/4$ both coefficients are nonnegative, so $\gamma(x,\delta) > 0$ at every finite δ. No relative rapidity inverts the force at a radius $r \geq 2\ell/\pi$.

For $\pi/4 < x < x_1$ one has $0 < \gamma_s < \sigma$, so the ratio $\rho = \gamma_s/\sigma$ lies in $(0,1)$. The factor (10) vanishes at

$$\tanh\delta = \rho(x),$$

and therefore at

$$\varphi_*(x) = 2\,\mathrm{artanh}\,\rho(x) = \ln\frac{\sin x + \cos x}{\sin x - \cos x} - 2x. \tag{12}$$

Below this threshold $\gamma > 0$ and the force is inward. Above it $\gamma < 0$ and the force is outward. As $x \downarrow \pi/4$, the denominator $\sin x - \cos x$ vanishes and $\varphi_* \to \infty$. As $x \uparrow x_1$, $\gamma_s \to 0$ and $\varphi_* \to 0$.

The threshold falls strictly with the phase. On $(0,\pi)$,

$$\gamma_s'\sigma - \gamma_s\sigma' = -2\sin^2 x. \tag{13}$$

The identity is the substitution of $\gamma_s' = -2\cosh x \sin x$ and $\sigma' = 2\sinh x \sin x$: the bracket that remains is $\cosh x\,\sigma + \sinh x\,\gamma_s$, and that bracket equals $\sin x$. Thus $(\gamma_s/\sigma)' < 0$ wherever $\sin x \neq 0$. In particular ρ falls from $+\infty$ to 0 on $(0, x_1)$, it equals 1 at exactly $\pi/4$, and φ_* falls from $+\infty$ to 0 on $(\pi/4, x_1)$. The radius $r_* = \ell/(2x_*)$ at which a given rapidity sits on the threshold therefore rises with φ. It starts at r_1 when $\varphi = 0$ and approaches $2\ell/\pi$ as $\varphi \to \infty$. The limiting ratio is

$$\frac{2\ell/\pi}{r_1} = \frac{4x_1}{\pi} = 1.1937. \tag{14}$$

The inversion layer thickens outward by less than one fifth, at any rapidity.

Inside the window the derivative of (10) with respect to δ is strictly negative. A higher rapidity lowers γ. While γ is still positive the inward force strengthens. At φ_* the inward force diverges. Immediately above φ_* the outward force diverges, and a further increase of φ then drives γ more negative, so the outward magnitude falls from infinity toward the ultrarelativistic size read off from the first term of (11). High relative velocity does not multiply a fixed repulsive law. It moves the sign change. The repulsive regime of a given separation in the window is the regime $\varphi > \varphi_*(x)$, and the repulsion is arbitrarily large at the entrance to that regime.

Outside the window the same derivative vanishes once, at $\tanh\delta = \sigma/\gamma_s < 1$. There γ has a positive minimum. A small rapidity strengthens the attraction, and a large rapidity weakens it toward zero, because (11) grows as $e^{\varphi/2}$. The force at $r > 2\ell/\pi$ remains inward at every finite speed.

At small φ the threshold lies close to the static node. The node condition is $\gamma_s(x_1) = 0$ with

$$\gamma_s'(x_1) = -2\cosh x_1 \sin x_1,$$

and (6) at a point where $\gamma_s = 0$ gives $\sigma(x_1) = \sin x_1/\cosh x_1$. The derivative of φ_* at the node is exactly $-4\cosh^2 x_1$, so

$$x_1 - x_* \sim \frac{\varphi}{4\cosh^2 x_1}, \qquad \frac{1}{4\cosh^2 x_1} = 0.11528. \tag{15}$$

To first order the fractional widening of the radius is this coefficient divided by x_1, which is $0.12295\,\varphi$. The phase gap itself is strictly smaller than $\varphi/(4\cosh^2 x_1)$, at every positive rapidity, and the fractional widening $(r_* - r_1)/r_1$ is strictly smaller than φ/π. The repulsive skin of a slow neutron is a neighbourhood of r_1 of that fractional thickness.

The map from a laboratory kinetic energy to φ uses the neutron rest energy $m_n c^2 = 939.6\,\mathrm{MeV}$ as an input, through $\cosh\varphi = 1 + T/(m_n c^2)$. The threshold radius at that rapidity is the unique r_* with $\varphi_*(\ell/(2r_*)) = \varphi$.

Table 2: Inversion radius of a neutron at kinetic energy T. The limiting ratio is $4x_1/\pi = 1.1937$.

T	φ	r_*/r_1
1 eV	4.6×10^{-5}	1.00001
1 keV	0.00146	1.00018
1 MeV	0.04613	1.00563
14 MeV	0.1724	1.0206
100 MeV	0.4574	1.0517
1 GeV	1.353	1.124

A thermal neutron, with φ of order 10^{-5}, is indistinguishable from the static node at the precision of the table. Across the whole window except that skin, the force on it is inward, and the inward force becomes large as the skin is approached. A neutron of 14 MeV meets the threshold only two percent of r_1 outside the static node. That energy is the energy of the neutron emitted by deuterium–tritium fusion. It is not the energy of a neutron that induces fusion, and the reaction which emits it is a charged entrance, treated below. At 100 MeV the shift is five percent. The geometry does not turn a few-MeV neutron back at twice the hard-core radius. It thickens the repulsive skin in proportion to φ, up to the strict wall $2\ell/\pi$.

At $\ell = \lambda_\pi \approx 1.41\,\mathrm{fm}$, that wall lies at $2\ell/\pi = 0.898\,\mathrm{fm}$. Read from the bounds $7/5 < \lambda_\pi < 71/50$, the same wall lies strictly between 8/9 and 142/157 fm. No finite rapidity moves the onset of repulsion beyond that wall. The static node sits at 0.752 fm, and the same bounds place it strictly between 14/19 and 213/280 fm. The ratio of the wall to the node is independent of the length.

The instantaneous criterion is not yet a capture cross section. While the force remains inward and the neutron is still moving inward, the rapidity itself increases. With the relative momentum $\mu c \sinh\varphi$ and with μ the reduced mass,

$$\mu c \cosh\varphi \, \frac{d\varphi}{dt} = \frac{\kappa}{\gamma r^2}, \qquad \frac{dr}{dt} = -c\tanh\varphi, \tag{16}$$

so long as $\gamma > 0$. Equivalently,

$$\sinh\varphi \, d\varphi = -\frac{\kappa}{\mu c^2}\,\frac{dr}{\gamma r^2}.$$

The right-hand side depends on φ through γ. While γ stays positive the rapidity increases as the separation falls, and on the window the threshold falls, so the margin $\varphi_* - \varphi$ narrows from both sides. It closes at a phase strictly below x_1. A nonnegative rapidity that increases from a value below the threshold meets the threshold once, before the node. The static node is not reached while the channel is still attractive. Reaching it on the attractive side requires the rapidity to fall. Radiation, excitation, and the imaginary part of an optical potential are not in (7), and the note does not derive them. It therefore does not integrate (16) through a compound nucleus and does not derive the laboratory rule that a capture cross section falls as $1/v$. What the signed force does fix is the surface on which the force changes sign, and the fact that a free inward rapidity meets that surface before the static node. A neutron that arrives at separation $r \in (r_1, 2\ell/\pi)$ with $\varphi > \varphi_*(\ell/(2r))$ is already on the outward side of a pole. A slower arrival at the same separation is on the inward side, and if its rapidity then keeps increasing it meets the pole before r_1.

The preference for a slow neutron belongs to the entrance, and the entrance is shared. Capture accepts the neutron and remains inward. Neutron-induced fission accepts the same neutron and may afterwards split. Fusion accepts a partner nucleus and remains inward. Above φ_* the channel is already outward, so none of the three begins. The formula is that common refusal. It does not choose which reaction follows once the partner is inside.

A charged projectile keeps this nuclear condition and adds the Coulomb term of the next section. Whether the inward nuclear term exceeds that Coulomb term is the charged entrance of Section 6. The projectile carries charge, so that entrance is not a slow-neutron reaction.

5 The chaotic proton force

A proton carries both channels. The gravitational channel is (7). The electromagnetic channel of a like pair, on its own equal-scale locus with its own length ℓ_E, contributes the signed Coulomb force of the companion note [2]. The force that increases the separation of two protons is

$$F_r = -\frac{\kappa}{\gamma(x,\varphi/2)\,r^2} + \frac{ke^2}{\gamma_s(x_E)\,r^2}, \qquad x = \frac{\ell}{2r}, \quad x_E = \frac{\ell_E}{2r}, \tag{17}$$

with $k = 1/(4\pi\epsilon_0)$. The rapidity is assigned to the inertial pair. The charge pair keeps the equal-scale factor. A neutron is the case $e = 0$.

Choice of length. Every particle carries both channels at once. An electron or a proton always exerts the gravitational force of its mass and the electric force of its charge; neither channel is ever switched off. What the rule chooses is the length that governs each channel's force. The gravitational channel is governed by λ_π for every mass: protons, neutrons, and electrons. The electron's gravitational force is the force of its mass. At the Bohr radius both interference factors are positive, and that force is smaller than the electric force between the electron and the proton by a factor strictly between 4.40×10^{-40} and 4.42×10^{-40}. The electromagnetic channel is governed by ℓ_E for charges that act as isolated charges, electrons and free protons, and by the common length λ_π for protons in proton chaos.

Whether a charge is isolated is decided by the electromagnetic channel itself, before either length is chosen for it. For any length ℓ, every separation beyond $2\ell/\pi$ lies on the outer plateau, where $\gamma_s > 0$ and no wall remains. A charge is isolated toward a partner when their separation lies there for $\ell_E \approx 386\,\mathrm{fm}$, beyond about 246 fm. The electron and proton of hydrogen, at the Bohr radius $a_0 \approx 5.29 \times 10^4$ fm, satisfy this, so the electric force between them is governed by ℓ_E, while their gravitational force acts as always at λ_π. Two protons of a nucleus do not. With ℓ_E, the window of separations from 1 to 2 fm would already carry at least thirty electromagnetic walls, and every separation below 193 fm, the head-on distance of an alpha particle on gold among them, lies inward of the first electromagnetic node. A nuclear proton is therefore not an isolated charge. Its gravitational channel with the other nucleons, neutrons included, is bounded by gravitational walls: separations of 1 fm or more lie beyond the wall $2\lambda_\pi/\pi$, in the outer attractive chamber, and a pair carried inward meets the wall, where its force changes sign. That sign cell, bounded by walls, is the proton chaos of this section. The chaos is reached without assuming which length governs the electromagnetic channel.

The rule then follows. The protons inside a nucleus are in proton chaos and do not act as isolated charges, so at the level of nuclei both channels act at λ_π: the gravitational channel as always, and the electromagnetic channel, the Coulomb term, at the common length. The choice is consistent with its own premise: at the common length the walls of the two-channel force are exactly the gravitational walls, and the sign-cell structure below is kept.

The two lengths are independent inputs. Write $\lambda = \ell_E/\ell$. The electromagnetic nodes, read in the gravitational phase, lie at x_n/λ. They coincide with the gravitational nodes for every n only when $\lambda = 1$. In that special case (17) at rest is a single layer sequence with prefactor $ke^2 - \kappa$. If $\kappa > ke^2$ the net force has the gravitational sign. If $\kappa < ke^2$ it has the Coulomb sign. There is no second set of walls. By the length rule, the proton pairs of an assembled nucleus are read in this case. The chambers are then the layers of the single sequence, which alternate between inward and outward, and the sign-cell reading of the chaotic proton force below applies to them unchanged. The interleaving for $\lambda \neq 1$ describes a pair in which a proton still acts as an isolated charge.

When $\lambda \neq 1$ the walls interleave. Each open interval that contains no node of either factor is a chamber: both denominators keep their signs, and the direction of each term is constant. The number of chambers that meet a bounded phase interval grows without bound as the inner endpoint is pushed

toward the origin, because each channel contributes one node per interval of length π. If λ is irrational, no gravitational node coincides with an electromagnetic node, every wall is a simple pole of only one term, and the sequence of chamber lengths in the phase is not periodic. The first wall met on the way in from large separation is the gravitational node when $\lambda < 1$, and the electromagnetic node when $\lambda > 1$. A finite Coulomb repulsion outside a shorter-range inversion is the first of these two orderings: $\ell_E < \ell$, the charge factor is still positive at r_1, and the gravitational pole overtakes that finite term in some neighbourhood of r_1. The width of the neighbourhood is set by $\kappa/(ke^2)$ and is not a pure number. The opposite ordering puts a Coulomb pole outside the gravitational node, and a classical approach from infinity is turned back there.

The stability of a wall is fixed by the sign in front of that term and by the sign of γ_s'. Consider a zero where γ_s falls through zero, so $\gamma_s' < 0$. On the gravitational side, both neighbouring chambers drive the separation toward the zero, as in Section 3. On the electromagnetic side the prefactor of $1/\gamma_s$ is positive rather than negative, and both neighbouring chambers drive the separation away from the zero: just outside, like charges repel and the separation grows; just inside, $\gamma_s < 0$ and like charges attract, so the separation falls and the phase moves away from the zero. A zero where γ_s rises through zero has the opposite stability in each channel. Consecutive walls of one channel therefore alternate between attractors and repellers of the separation. The outermost gravitational wall is an attractor. The outermost electromagnetic wall is a repeller.

The interleaving is already visible for the illustration $\lambda = 2$, which is not a determined value of the ratio. As the gravitational phase x increases from zero, the first wall is the electromagnetic node at $x = x_1/2 = 0.46878$. Beyond it, and until the gravitational node at $x_1 = 0.93755$, the electromagnetic factor is negative, so the Coulomb term is inward, and the gravitational term is still inward. That chamber pulls from both channels. Past x_1, and until the next electromagnetic node at $x = x_2/2 = 1.9637$, the gravitational term has become outward while the Coulomb term remains inward. The net direction is then the comparison

$$\frac{\kappa}{|\gamma|} \gtrless \frac{ke^2}{|\gamma_E|}, \tag{18}$$

which depends on the input κ and on the two denominators. Each later wall reverses exactly one of the two terms. A proton pair that moved inward through this sequence would alternate between a doubly inward chamber, a contested chamber, and chambers in which the Coulomb term has flipped again. Classically the pair is not free to cross an attracting wall: the force diverges there. The alternation is the chamber structure, not a smooth well with a single minimum.

In a nucleus the pairs do not share one separation. Consider three protons on a line, at 0, s, and $s+t$, with s, t, and $s+t$ off every node. Let $F(s)$ denote (17) at separation s, positive when it increases that separation. The rightward force on the middle proton is $F(s) - F(t)$. If the left separation lies in an inward chamber, $F(s) < 0$, and the right separation lies in an outward chamber, $F(t) > 0$, the difference is negative: the left partner pulls the middle proton leftward, and the right partner pushes it leftward. The two forces point the same way. A displacement that carries one of those separations onto a node sends the corresponding F through infinity and reverses it. The change in the force on the middle proton is unbounded, at an arbitrarily small displacement of the wall crossing. Configurations on the two sides of the wall are not close in force.

That is the chaotic proton force. It is not a claim that the radial two-body problem is a chaotic dynamical system: one separation, with a prescribed $F(r)$, is a single degree of freedom. The chaos meant here is the arrangement. Several pairs sit in several chambers at once, some inward and some outward, and the vector sum on each proton is smooth only in the interior of a sign cell. The boundary of the cell is a pole. A nucleus is not a collection of pairs in one attractive well. It is a sign cell of the pair separations, stable only so long as no pair meets a wall, and sensitive to a displacement that brings one pair onto a wall.

The Newtonian identification makes the contest (18) extreme away from the walls. The CODATA values give

$$\frac{Gm_p^2}{ke^2} = 8.094 \times 10^{-37}, \qquad \frac{ke^2}{Gm_p^2} = 1.236 \times 10^{36} \tag{19}$$

[5]. If κ is replaced by Gm_p^2, the gravitational term exceeds the Coulomb term only where $|\gamma|$ is smaller than $|\gamma_E|$ by that factor 1.236×10^{36}. On a plateau, where both factors are of order one or larger, the Coulomb term decides the sign. The main text already refuses to conflate $Gm_p^2/\hbar c \sim 10^{-38}$ with an $O(1)$ nuclear strength [1]. The refusal is kept. The prefactor κ of the nuclear channel is an input. The chamber walls, their stability, and the rapidity threshold do not depend on it. Which term wins inside a contested chamber does.

A proton–neutron pair has the gravitational term and no Coulomb term. A proton–proton pair has both. The outer gravitational well can bind the first and fail to bind the second once ke^2/γ_E exceeds κ/γ, which is the uncontested direction of (18) wherever both factors are positive. The laboratory deuteron is a proton–neutron bound state, and the diproton is unbound. The note does not compute the 2.2 MeV binding, and it does not derive the neutron–neutron scattering length: two neutrons at a separation outside the node are not on the pole that excludes a shared seat, so the sign of γ_s alone does not decide that they are unbound.

6 The slow charged entrance

The Coulomb term does not by itself decide whether a charged pair can enter. Write $\rho = ke^2/\kappa$, and write the force that increases the separation as

$$F_r = \frac{\kappa}{r^2}\left(-\frac{1}{\gamma} + \frac{\rho}{\gamma_E}\right), \tag{20}$$

wherever both denominators are nonzero. The sign of the force is the sign of the second factor. A charged pair at the entrance is a pair of nuclei. Both channels act on it, and by the length rule of Section 5 both are governed by the common length $\ell_E = \ell = \lambda_\pi$. The rapidity belongs to the inertial pair, so the electromagnetic factor is the static equal-scale factor at the same phase, $\gamma_E = \gamma_s(x)$. On the entrance window $\pi/4 < x < x_1$ that factor is strictly positive. The positivity of γ_E used throughout this section therefore holds on the window itself, and no ordering of two different lengths is needed. The gravitational node is the first wall met from large separation, and at that node the Coulomb term is still finite.

On the window $\pi/4 < x < x_1$ the interference factor falls strictly as the half-rapidity $\delta = \varphi/2$ rises. Above the threshold, γ is negative, so $-1/\gamma$ is positive and the nuclear term adds to the Coulomb term. The net coefficient is then strictly larger than ρ/γ_E. The repulsion exceeds the Coulomb repulsion, and it is arbitrarily large immediately above the threshold. A rapidity raised in order to cross a Coulomb barrier meets, inside the window, a nuclear repulsion that has joined the barrier.

Below the threshold, γ is positive and falls to zero at the threshold, so $-1/\gamma$ falls without a lower bound. For every finite positive ρ and every positive γ_E there is an interval of rapidity, abutting the threshold from below, on which the net coefficient is negative. The nuclear term exceeds the Coulomb term there.

The threshold phase moves toward the node as the rapidity falls, so a smaller positive rapidity remains inward down to a smaller radius. The phase gap and the fractional widening are those already fixed for the slow-neutron skin. Whenever $\cosh\varphi = 1 + T/(mc^2)$ with

$$0 < \frac{T}{mc^2} \le 10^{-10},$$

that widening is strictly less than 5×10^{-6}. A nucleon or a deuteron at room temperature is of this kind. The thermal energy is a few hundredths of an electron-volt and the rest energy is of order a GeV, so the ratio lies below 10^{-10}. The repulsive skin is then thinner than the skin already invisible, at the precision of Table 2, for a neutron of 1 eV. At such a rapidity the onset of nuclear repulsion is indistinguishable from the static node, and the neighbourhood in which the nuclear term exceeds the Coulomb term sits against that node.

Outside the radius $2\ell/\pi$ the nuclear term does not change sign at any finite rapidity. The interference factor grows without bound as the rapidity grows, the nuclear attraction tends to zero, and beyond some rapidity the net force is outward: the pair is left to the Coulomb repulsion. Slowing, at a separation outside that radius, restores a nuclear attraction that a fast passage has erased. The rapidity of strongest attraction on that side remains the one already located, where the derivative in the rapidity vanishes.

What favours a low temperature is the side of the pole and the place of the pole. Below the threshold the inward coefficient becomes more negative as the rapidity rises, so the advantage is not a stronger pull at rest.

The free infall of Section 4 still raises the rapidity, and a rising rapidity still meets the pole before the node. A slow charged pair that is then left to fall freely is turned away on that pole. The low-temperature entrance is the entrance in which the rapidity is kept below the falling threshold while the separation approaches the node. On the axis of a regular octahedron of positive ions, the electromagnetic channel holds two like charges at a restoring seat outside the first charge node, and that seat keeps the separation off a free electromagnetic infall through the charge node [3]. At the reduced Compton wavelength of the

electron, the cage radius 2ℓ of that seat lies strictly between 772.2 fm and 772.4 fm, while the octahedral hole of palladium is about 2 Å, larger by a factor of about 250. The force does not derive a rate, an excess of heat, or a choice among the experiments that have asked whether a cold charged entrance occurs. It fixes the sign. On a path that stays below the threshold, the net force is inward throughout some interval that ends at the threshold, for every finite ρ, so long as γ_E stays positive. The positivity holds on the whole window at the common length, so no Coulomb pole stands outside the interval. Both channels also act at the seat in the octahedral hole. Two charges about 2 Å apart lie beyond $2\ell_E/\pi$, on the wall-free plateau of the electromagnetic length, and act there as isolated charges: their electric force is governed by ℓ_E, their gravitational force by λ_π. Along the approach from the seat to the nucleus only the length governing the electromagnetic channel changes, from ℓ_E to the common length. Where along the approach that change takes place is under study.

7 Magic numbers and nuclides

The node radii scale as $\Theta(1/n)$. That sequence does not contain the integers 2, 8, 20, 28, 50, 82, and 126. Those integers count seats, in the same sense in which the electromagnetic companion reads 2, 8, 18, and 32 as two chiral seats on successive angular degrees and refuses to read them as node labels [2]. The nuclear count is a different arithmetic, because the empirical closures split the oscillator shells. The splitting is not derived from γ_s. What follows is the count itself, so that it is not mistaken for a property of the inversion layer.

The three-dimensional isotropic oscillator shell $N = 0, 1, 2, \ldots$ has orbital degeneracy $(N+1)(N+2)/2$. Two spin seats on each orbital give $(N+1)(N+2)$ states. The cumulative number through shell N is

$$C(N) = \sum_{k=0}^{N} (k+1)(k+2) = \frac{(N+1)(N+2)(N+3)}{3}. \tag{21}$$

The first values are $C(0) = 2$, $C(1) = 8$, $C(2) = 20$, $C(3) = 40$, $C(4) = 70$, and $C(5) = 112$. Shell $N+1$ has highest orbital angular momentum $l = N+1$, and the state $j = l + 1/2$ has degeneracy $2j + 1 = 2(N+2)$. Adding that one orbital to the cumulative count produces

$$C(N) + 2(N+2) \tag{22}$$

for $N \geq 2$, namely 28, 50, 82, and 126. The observed closures are $C(0)$, $C(1)$, $C(2)$, and (22). The pure oscillator closures $C(N)$ for $N \geq 3$ are not among them. The arithmetic is the shell-model arithmetic: one high-j orbital of the next shell is counted with the closed shells below it. The layer geometry does not say why that orbital, rather than another, is the one that moves, and this note does not supply the missing reason. Charge independence and spin–orbit coupling remain, as in the main text, resonance conditions of the dual rotor that have not been reduced to the nodes.

Proton seats and neutron seats are seats of two rotor classes [4]. Each class has the count above. The electromagnetic term and the chaotic sign cell of Section 5 act in the proton class. They do not act in the neutron class. The integers can therefore agree while the binding does not. Adding a proton to a configuration adds a Coulomb sum over the proton pairs already present, in every chamber where $\gamma_E > 0$, and it adds one further proton pair to the sign cell. Adding a neutron adds neither. The stable composition of a heavy nuclide bends toward neutron excess for that reason. The coefficient of $Z^2/A^{1/3}$ in the semi-empirical mass formula is not fixed by the bend, and it is not derived here.

A doubly magic nuclide is one whose proton count and neutron count are each one of 2, 8, 20, 28, 50, 82, 126.

Among these, ^{4}He, ^{16}O, ^{40}Ca, ^{48}Ca, and ^{208}Pb are stable. The others lie off the valley. The seat count closes in either case. The maximum of BE/A lies near $A = 62$, between the closures 28 and 50, rather than on ^{56}Ni. The Coulomb term on the proton class is available as the agent that moves the maximum off the doubly closed point. The value of the maximum remains the input already quoted.

The discrete theory proves $A_{\max}(N) < \infty$ at each fixed compactification and does not evaluate it [4]. The estimate $A_* \sim 300$ is the heuristic of that paper and of the main text, read off from the empirical end of the magic sequence together with a layer budget. The nodes of Section 3 do not tighten it. If ℓ is independent of A, the nuclear surface of the liquid-drop radius already lies outside every node at $A = 1$. If instead ℓ scales as $A^{1/3}$, every ratio of the surface to a node is independent of A, and again no cutoff appears. Finiteness of the nuclide chart, in this framework, is the finiteness of the rotor image, not a property of γ_s.

Table 3: Doubly closed counts. Closure of the two counts is not stability against decay.

Nuclide	Z	N	A
^{4}He	2	2	4
^{16}O	8	8	16
^{40}Ca	20	20	40
^{48}Ca	20	28	48
^{56}Ni	28	28	56
^{78}Ni	28	50	78
^{100}Sn	50	50	100
^{132}Sn	50	82	132
^{208}Pb	82	126	208

8 The shared entrance, and fission

A dual excitation is an addition to the dual angle at fixed proximity, the complementary reading to (9):

$$\alpha = \frac{\ell}{r}, \qquad \beta = \frac{\ell}{r} + \psi, \qquad \psi \geq 0. \tag{23}$$

With $\varepsilon = \psi/2$ and with τ as in (5),

$$\gamma_\psi(x) = \cos\varepsilon\, \gamma_s(x) - \sin\varepsilon\, \tau(x). \tag{24}$$

On the outer well $0 < x < x_1$, both γ_s and τ are positive. For $\psi \in (0, \pi)$ one has $\varepsilon \in (0, \pi/2)$ and $\cos\varepsilon > 0$, so γ_ψ has the sign of $\gamma_s/\tau - \tan\varepsilon$. The zero lies at

$$\psi_*(x) = 2 \arctan \frac{\gamma_s(x)}{\tau(x)}. \tag{25}$$

The quotient γ_s/τ is strictly decreasing on $(0, x_1)$. The numerator of its derivative is $\gamma_s'\tau - \gamma_s\tau'$. Substituting $\gamma_s' = -2\cosh x \sin x$ and $\tau' = 2\cosh x \cos x$ leaves the bracket $\sin x\, \tau + \gamma_s \cos x$, and that bracket equals $\cosh x$, so the numerator is $-2\cosh^2 x$. Thus ψ_* falls strictly from π to 0 as x runs from 0 to x_1. Every excitation $\psi \in (0, \pi)$ is the critical value at exactly one phase x_ψ in the outer well.

At small phase, $\tau(x) = 2x + O(x^3)$ and $\gamma_s(x) = 1 + O(x^2)$, so

$$\psi_*(x) = \pi - 4x + O(x^3) = \pi - \frac{2\ell}{r} + O\Big(\Big(\frac{\ell}{r}\Big)^3\Big).$$

A widely separated pair is inverted only by an excitation near π. A pair already near the node is inverted by a small excitation. Relative rapidity cannot imitate this at large separation: outside $2\ell/\pi$ the factor (10) stays positive at every finite φ.

The phase x_ψ is an attractor of the separation at fixed ψ. For $x > x_\psi$ one has $\psi_*(x) < \psi$, hence $\gamma_\psi < 0$, and the force is outward, so the separation grows and the phase falls toward x_ψ. For $x < x_\psi$ the force is inward, the separation shrinks, and the phase rises toward x_ψ. The zero of γ_ψ is simple because $(\gamma_s/\tau)' \neq 0$, and the force has a simple pole there. As $\psi \downarrow 0$ the attractor approaches the static node. As $\psi \uparrow \pi$ one has $x_\psi \downarrow 0$, so the attractor recedes to infinity. From the leading deficit, the radius of the attractor at an excitation close to π is $2\ell/(\pi - \psi)$.

The rapidity threshold and the dual threshold are not two names for one reaction. The rapidity threshold φ_* is the entrance, and it is common. Capture, neutron-induced fission, and fusion each require a partner that is still on the inward side of φ_*. Above that threshold the nuclear channel repels, and the partner is turned away before a compound system exists. Fusion is this entrance completed: the combined system remains inward, and the mass number has increased. For two charged nuclei the Coulomb term of Section 5 remains, and the comparison of the two terms is the entrance of Section 6. A neutron already carries no charge. Its being slow is what keeps φ under φ_*.

Fission is not that entrance. It is a dual excitation of a nucleus already formed, not a high relative rapidity and not the joining itself. The main text describes it as an excited dual mode that drives γ_s from positive to negative and repels the fragments [1]. The formula (25) locates that transition. On the inner side of x_ψ the force is outward, which is the repulsion of the fragments. On the outer side the outer well is still attractive, so a fixed excitation gathers the separation at $r = \ell/(2x_\psi)$ rather than sending it to infinity. A split that runs away requires the excitation to approach π, where the gathering

radius diverges, or requires the excitation to change during the separation. Neither the Q value nor the asymmetric mass split of actinide fission is computed. Neutron-induced fission uses both thresholds in that order. The neutron arrives with φ below φ_*, or it never joins. The excitation ψ deposited by that joining is what can cross ψ_*. Spontaneous fission has no neutron entrance. It is only the second threshold. The branching between capture, scattering, fusion, and fission is not decided by the sign of γ alone.

The two thresholds therefore answer two different questions, and a slow neutron sits in both only as the way in. The rapidity threshold says when a moving nucleon, at a separation already inside $2\ell/\pi$, finds the channel outward. The dual threshold says when an internal excitation inverts a pair at rest, at any separation. Joining uses the first. Splitting uses the second.

9 Rutherford scattering

An alpha particle on gold meets the Coulomb term of (17) at separations of tens of femtometres. Rutherford's law, $d\sigma/d\Omega = (d/4)^2/\sin^4(\theta/2)$ with $d = 2Zke^2/E$, is the law of an unsigned Coulomb repulsion down to the head-on distance d [8, 9]. For the inputs $Z = 79$, $E = 7.69\,\mathrm{MeV}$, and $ke^2 = 1.439964\,\mathrm{MeV\,fm}$, the laboratory-frame distance d lies strictly between 29 and 30 fm.

By the length rule of Section 5, the alpha particle and the gold nucleus interact as bodies of nucleons, whose protons are in proton chaos, and the length read between them is λ_π, for the Coulomb term of (17) as well. This is the case $\ell_E = \ell = \lambda_\pi$. The head-on distance exceeds $20\lambda_\pi$ and lies outside the first node. At every separation of at least 29 fm the signed factor exceeds one by less than six parts in ten thousand, and the departure grows as $(\lambda_\pi/2r)^2$.

In the centre-of-mass frame the kinetic energy is the laboratory energy times $A/(A+4)$, with A the target mass number, so the head-on distance is multiplied by $(A+4)/A$. The departure there is 5.5×10^{-4}. Classical deflections of the signed force then give the following excess of the cross section over Rutherford's. At 7.69 MeV it is 3.1×10^{-4} at 90° and 5.8×10^{-4} at 170°. At 5.3 MeV it is 1.5×10^{-4} and 2.8×10^{-4}. At 2 MeV, an energy of backscattering analysis, it stays below 4×10^{-5}. These excesses lie far below the percent at which large-angle data are compared with Rutherford's law. The potential of that force, integrated from the contact radius $1.2(A_d^{1/3}+4^{1/3})$ fm of the daughter and the alpha particle out to the turning point, raises the Gamow exponent of ^{212}Po by 0.037 and that of ^{238}U by 0.033. The half-lives rise by 3.8 and 3.4 percent, far inside the spread of the preformation factor.

Anomalous scattering of alpha particles on heavy nuclei sets in when the two surfaces approach, near $1.2\,(A_1^{1/3}+A_2^{1/3})\,\mathrm{fm} \approx 9\,\mathrm{fm}$ for gold. At 10 fm the factor departs from one by 5×10^{-3}. In this reading a departure of nuclear size begins when nucleon pairs across the gap come within a few λ_π and approach the node, each at its own separation. That is the place where the data depart. The formula (17) is a two-body force and does not sum those pairs.

The nuclear prefactor κ is a pair input, and this note does not fix how it adds over the nucleons of two nuclei. If it adds coherently over the A_1A_2 pairs, the far field has prefactor $Z_1Z_2ke^2 - A_1A_2\kappa$, and the cross section is scaled by $(1-\eta)^2$, with $\eta = A_1A_2\kappa/(Z_1Z_2ke^2)$. For alpha on gold $A_1A_2/(Z_1Z_2)$ lies strictly between 4.98 and 4.99. A far-field share of at most 1/200, a change of one percent in the cross section, then forces $\kappa < ke^2/997$. The Newtonian prefactor Gm_p^2 gives $\eta \approx 4\times10^{-36}$. The length does not scale with the mass number here: λ_π is the length of a nucleon pair.

The growth of the cross section with the nuclear charge is the growth of the Coulomb term. Rutherford's law scales as Z^2. From alpha scattering on copper, silver, and platinum, Chadwick obtained the nuclear charges 29.3, 46.3, and 77.4, against the atomic numbers 29, 47, and 78, with errors of about one percent [10]. Under the coherent sum the effective charge is $Z(1-\eta)$, with $\eta = 2(A/Z)\,\kappa/(ke^2)$ for an alpha particle, and the coefficient $2A/Z$ lies between 4.38 and 4.39 for copper, between 4.59 and 4.60 for silver, and between 5.00 and 5.01 for platinum. A least-squares fit of the three charges, with one percent on each, gives $\kappa/(ke^2) = (0.94 \pm 1.24)\times10^{-3}$, consistent with zero. The gravitational term scales with the mass numbers, and the measured growth follows the charge. The signed Coulomb factor at λ_π adds a small excess that falls with the charge, because a lighter nucleus is approached more closely. At 7.69 MeV and 150° in the same frame the excess is 1.6×10^{-2} for aluminium, 3.8×10^{-3} for copper, 1.5×10^{-3} for silver, and 5.5×10^{-4} for gold. For aluminium that path reaches 5.7 fm, at the contact radius $1.2\,(A_1^{1/3}+A_2^{1/3})\,\mathrm{fm} \approx 5.5\,\mathrm{fm}$, where the two-body formula is not the whole account.

On gold, the separations of this scattering stay within that bound. The open input is the sum of κ over nucleons.

10 Neutron scattering

A neutron carries both channels. Its charge is zero, so the electric term is absent, and its force on a nucleon is (7) at the length λ_π, with the relative rapidity of Section 4, and its repulsion changes with its speed through (10). The onset of repulsion lies at the static node $r_1 = 0.7540\,\mathrm{fm}$ for a slow pair and moves outward as the rapidity grows. For a thermal neutron of $0.0253\,\mathrm{eV}$ the rapidity is 7.34×10^{-6}, and the onset moves by 9 parts in ten million. At $1\,\mathrm{MeV}$ the rapidity is 0.0461, and the onset lies at $0.7582\,\mathrm{fm}$. At $14\,\mathrm{MeV}$ the rapidity is 0.172, and the onset lies at $0.7695\,\mathrm{fm}$, two percent outside the node. At every speed it stays inside the wall $2\lambda_\pi/\pi = 0.9001\,\mathrm{fm}$. Each nucleon pair therefore carries a repulsive skin whose thickness grows with the energy of the neutron, and that energy dependence is a property of the gravitational channel itself.

The measured scattering is not the same for every nucleus. The bound coherent scattering lengths of thermal neutrons are $3.449\,\mathrm{fm}$ for aluminium, 7.718 for copper, 5.922 for silver, 7.63 for gold, and 9.405 for lead, and they are negative for hydrogen, -3.739, titanium, -3.438, and vanadium, $-0.3824\,\mathrm{fm}$ [11]. The total bound scattering cross sections run from $1.503\,\mathrm{b}$ for aluminium to $11.118\,\mathrm{b}$ for lead and $18.5\,\mathrm{b}$ for nickel, without a monotone order in A. At $14\,\mathrm{MeV}$ the evaluated total cross sections are $1.744\,\mathrm{b}$ for ^{27}Al, 2.909 for ^{63}Cu, 2.959 for ^{65}Cu, 4.258 for ^{107}Ag, 4.396 for ^{109}Ag, 5.295 for ^{197}Au, and $5.385\,\mathrm{b}$ for ^{208}Pb [12]. The geometric form $2\pi(r_0 A^{1/3} + \hbar/p)^2$ reproduces these $14\,\mathrm{MeV}$ cross sections with r_0 between 1.35 and $1.50\,\mathrm{fm}$, an interval that contains λ_π. At $14\,\mathrm{MeV}$, $\hbar/p$ is $1.2\,\mathrm{fm}$. This note does not derive that agreement.

Between these regimes the reduced wavelength falls from $2.9 \times 10^4\,\mathrm{fm}$ at thermal energy to that $1.2\,\mathrm{fm}$, together with the resonances of each nucleus. For gold the evaluated total falls from $5.755\,\mathrm{b}$ at $1\,\mathrm{MeV}$ to $5.295\,\mathrm{b}$ at $14\,\mathrm{MeV}$, while the skin of each pair widens by one and a half percent. The skin is a refinement on the scale of the nucleon pair, and the size of the nucleus and the reduced wavelength carry the larger part of the measured change. The far field is a separate correction, and it is fixed. At vanishing phase the interference factor of a pair equals $\cosh(\varphi/2)$, so the force is the inverse-square law of strength $\kappa/\cosh(\varphi/2)$. With the neutron rest energy $939.565\,\mathrm{MeV}$, the half-angle reduction writes that strength as $\kappa/\sqrt{1 + T/(2m_n c^2)}$. At $1\,\mathrm{MeV}$ the square root lies strictly between 1.000266 and 1.000267. At $14\,\mathrm{MeV}$ it lies strictly between 1.003718 and 1.003719. From the first of these energies to the second, the coefficient falls by a fraction strictly less than 4/1000. That fall is smaller than the widening of the skin. The cross section of the point field has a separate energy dependence. An inverse-square attraction of strength $A\kappa/\cosh(\varphi/2)$ is a Rutherford field, and at a fixed angle its cross section falls as $1/(T^2(1 + T/(2m_n c^2)))$. From $1\,\mathrm{MeV}$ to $14\,\mathrm{MeV}$ that factor lies strictly between 197 and 198. Over the same interval the evaluated total for gold remains strictly above nine tenths of its value at $1\,\mathrm{MeV}$. The measured total is not this peak.

At every phase $x \in (0, \pi/4)$ and at every finite rapidity the force remains inward, and it exceeds this inverse-square law by a fraction smaller than $e^{-x}/(\cos x - \sin x) - 1$. When the phase is at most 7/100, the fraction is strictly less than 6/1000.

While the neutron stays outside the rapidity wall of every nucleon, the sum of the pair forces does not turn it away. The nucleons lie in some ball. If the neutron lies outside the concentric ball enlarged by $2\lambda_\pi/\pi$, every pair is outside its wall. Each pair force then points toward its nucleon, and the net force has a negative radial component. No arrangement inside the ball, and no finite speed, reverses that component before a wall is reached. Farther out, the sum lies close to the point field. Let the nucleons lie in a ball of radius R and let D be the distance from the neutron to the centre, with $D > R$. If the phase at distance $D - R$ is at most 7/100, the net force differs from the point field of strength $A\kappa/\cosh(\varphi/2)$ by a fraction of that field strictly less than

$$\frac{6}{1000} + \left(1 + \frac{6}{1000}\right) \frac{2(R/D)}{(1 - R/D)^3}.$$

When D is at least $50R$, the fraction is strictly less than 1/20. In that region an appreciable κ adds a forward peak of Rutherford's form, scaled by A^2. The Newtonian prefactor Gm_p^2 makes the peak negligible. The speed-dependent skin is not in this field. With the dual phase held at zero it begins only when some pair enters its own wall. On that locus the far-field factor equals $\cosh(\varphi/2)$ and is at least one, so the rapidity weakens the attraction and does not reverse it. Every nucleus then presents the same coefficient.

A sample need not share that phase. At vanishing separation phase one has $\alpha = \varphi$ and $\beta = \psi$, and (1) becomes

$$\cosh\frac{\varphi}{2}\cos\frac{\psi}{2} - \sinh\frac{\varphi}{2}\sin\frac{\psi}{2}.$$

The rapidity belongs to the beam and is common. The relative dual phase ψ belongs to the nucleus, and the far-field coefficient κ/γ fluctuates with it. Past the right angle the factor is negative at every nonnegative rapidity, and that nucleus repels. Short of the right angle the factor stays positive when the half-rapidity does not exceed the angular gap. At zero rapidity the factor vanishes on the right angle, and on either side it can lie closer to zero than any positive bound, so the strength $\kappa/|\gamma|$ exceeds every bound. A rapidity alone does neither.

The coherent coefficient of the sample is the weighted mean of these coefficients. The mean-square departure from the mean is nonnegative, and it vanishes when, and only when, every nucleus that occurs presents the same coefficient. Where two coefficients of opposite sign both occur, the mean lies strictly between them and is smaller in magnitude than the larger one. A negative coherent length can therefore sit over a sample in which some nuclei still attract.

The values in femtometres are external. Natural hydrogen has bound coherent and incoherent cross sections 1.7568 b and 80.26 b, and the second exceeds the first by more than forty-five. Natural vanadium has 0.0184 b and 5.08 b, and the second exceeds the first by more than two hundred. The even isotopes ^{46}Ti and ^{48}Ti are each listed with a vanishing incoherent cross section and with bound coherent lengths 4.93 fm and -6.08 fm, so the sign belongs to the nuclide. Weighting the five stable lengths 4.93, 3.63, -6.08, 1.04, and 6.18 fm by the abundances 8.2%, 7.4%, 73.8%, 5.4%, and 5.2% puts the mean strictly between -3.44 and -3.43 fm, beside the natural length -3.438 fm. Carbon-12 is listed with a vanishing incoherent cross section: one length, and no fluctuation. The skin is not required for the fluctuation.

11 Conclusions

The repulsive inversion layer is the first interval on which the equal-scale factor γ_s is negative. Its outer wall is the node $x_1 = 0.93755$, at $r_1/\ell = 0.53330$, and its inner wall is $r_2/\ell = 0.12731$. The outer wall attracts the separation from both sides. On the equal-scale locus a separation is driven toward that wall and not through the layer. A free infall meets the rapidity threshold strictly outside the wall and does not enter the layer. The plateau value of the outward force inside the layer, at the minimum of γ_s, is $3.4057\,\kappa/\ell^2$. The prefactor κ is a nuclear input. The combination Gm_p^2 is smaller than ke^2 by 1.236×10^{36} and is not that input. The length is the charged-pion Compton wavelength λ_π. The laboratory signs place it strictly between 56/75 fm and 19/10 fm, and its first node lies strictly between the core scale $\frac{1}{2}$ fm and the attractive radius 1 fm.

Added to the usual proximity rapidity, a relative rapidity φ lowers the interference factor. The force at phase $x \in (\pi/4, x_1)$ is inward for $\varphi < \varphi_*(x)$ and outward for $\varphi > \varphi_*(x)$, with φ_* the closed logarithm (12). No finite rapidity inverts the force outside $2\ell/\pi$, and the outward displacement of the wall is bounded by the ratio 1.1937. The repulsive skin of a slow neutron has fractional thickness $0.12295\,\varphi$. At 1 MeV the wall has moved by six parts in a thousand; at 100 MeV, by five percent. A neutron on the inward side of its own threshold is pulled in. A neutron on the outward side is driven off a pole. If the inward rapidity keeps increasing, the pull ends on the threshold, strictly outside the static node. That alternative is the entrance shared by capture, by neutron-induced fission, and by fusion. Shedding of the rapidity, and the $1/v$ law, are not derived.

Two protons add the signed Coulomb force. Walls at which γ_s falls through zero attract a mass pair and repel a like charge pair, and consecutive walls of either channel have the opposite stability. Unequal lengths interleave the walls, so a nuclear configuration is a sign cell: inward pairs and outward pairs at one instant, with an unbounded force on each wall. That cell is the chaotic proton force. Equal lengths produce no such interleaving, only the single prefactor $ke^2 - \kappa$.

The closures 2, 8, 20 and 28, 50, 82, 126 are the cumulative oscillator counts $C(N)$ and the shifted counts $C(N) + 2(N + 2)$. They label seats of each rotor class, not nodes. Doubly magic nuclides close both classes. Stability, the valley, and the maximum of BE/A near $A = 62$ are not consequences of the closure, and $A_{\max}$ remains the finite but unevaluated bound of the discrete theory.

A dual excitation ψ inverts the outer well at a single phase, and that phase attracts the separation. The critical angle falls from π at infinity to 0 at the first node. Fission is this inversion of a nucleus already formed. Fusion is not that inversion: it is the shared entrance, $\varphi < \varphi_*$, completed while the combined system is still inward. A slow neutron induces fission only by passing that entrance. The split, if it follows, is $\psi > \psi_*$. Relative rapidity does not produce the split outside $2\ell/\pi$.

For a charged pair the same entrance is the slow side of the pole. Above φ_* the nuclear term adds to the Coulomb term, and the net repulsion exceeds the Coulomb repulsion. Below φ_* the nuclear term opposes it, and immediately below the threshold the opposition exceeds every finite ratio ke^2/κ, so long as the electromagnetic factor stays positive. The strongest inward force on that side sits next to the pole

rather than at rest. A lower rapidity moves the pole closer to the node, and a kinetic energy at most 10^{-10} of the rest energy widens the radius by less than 5×10^{-6}. Room temperature for a nucleon or a deuteron is in that class. Outside $2\ell/\pi$ a large rapidity erases the nuclear attraction and leaves the Coulomb repulsion. The free infall still meets the pole before the node, so the cold entrance is a path on which the rapidity is held down. The host of that holding, along one axis, is the restoring seat outside the first charge node in a regular octahedron of positive ions [3]. The rate is not derived. The strong force, throughout, is the sign of one interference factor: inward on the outer side of the first node, outward in the inversion layer, inward again only in a deeper chamber that a classical infall does not reach, and outward under either a rapidity past φ_* or a dual excitation past ψ_*.

A neutron outside the rapidity wall of every nucleon is pulled inward when the relative dual phase vanishes. The far field of each pair is then the inverse-square law of strength $\kappa/\cosh(\varphi/2)$, exceeded by less than 6/1000 when the phase is at most 7/100. If the nucleons lie in a ball of radius R and the neutron is at a distance $D \geq 50R$ from the centre, with the phase at $D - R$ at most 7/100, the net force differs from the point field of strength $A\kappa/\cosh(\varphi/2)$ by a fraction strictly less than 1/20. The cross section of that point field, at a fixed angle, falls from 1 MeV to 14 MeV by a factor strictly between 197 and 198, while the evaluated gold total remains above nine tenths of its value. The measured total is not that peak. With the dual phase held at zero the skin meets the neutron only inside a wall, and the rapidity does not reverse the far field. The fluctuation from nucleus to nucleus is the fluctuation of the relative dual phase. It changes the far-field factor, including its sign, and the mean-square departure vanishes only when every nucleus in the sample presents the same factor. The values in femtometres are not derived.

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